SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Analytical Problem Solving

Code of Conduct:

I Moshitha kolanjiyappan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Moshitha k

Annexure A

Short Answer Test

1, A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is –0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails. (5 Marks)

Given:

- $w1 = 0.6$, $w2 = 0.4$
- Bias $b = -0.5$
- $x1 = 2$, $x2 = 1$

Step 1: Net Input

$Net = (w1x1) + (w2x2) + b$

$= (0.6 \times 2) + (0.4 \times 1) - 0.5$

$= 1.2 + 0.4 - 0.5 = 1.1$

Step 2: Step Activation

$f(x) = \{1, Net \geq 0$

$0, Net < 0$

Since $1.1 > 0$

Output = 1**Conclusion**

Student Passes.

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training. (5 Marks)

Given

- Weight = 0.5
- Learning rate $\eta = 0.1$
- Input $x = 2$
- Target $T = 1$

• Actual Output O = 0.6

Error

$Error = T - O = 1 - 0.6 = 0.4$

Weight Update

$\Delta w = \eta \times Error \times x$

$= 0.1 \times 0.4 \times 2 = 0.08$

Updated Weight

$wnew = 0.5 + 0.08 = 0.58$

Answer

• Error = **0.4**

• Weight Update = **0.08**

• New Weight = **0.58**

3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors. (5 Marks)

3. Confusion Matrix

Given

• TP =180

• TN =150

• FP =20

• FN =50

Accuracy = Accuracy=TP+TN/ FP+FN+TP+TN×100

Calculation: Accuracy = (180 + 150) / (180 + 150 + 20 + 50) × 100 = 330 / 400 × 100 = 82.5% Answer: Accuracy = 82.5%

4.A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms. (5 Marks)

Given:

Objective Function:

$$f(x) = x^2$$

Initial Population:

2, 4, 6, 8

Step 1: Calculate Fitness Values

Individual (x)	Fitness $f(x) = x^2$
2	4
4	16
6	36
8	64

Step 2: Select the Best Individuals

The individuals with the highest fitness values are:

- **8 (Fitness = 64)**
- **6 (Fitness = 36)**

These are selected as the parent chromosomes.

Step 3: Crossover

Binary representation:

- 8 = **1000**
- 6 = **0110**

Exchange the least significant bit (LSB):

Parent 1: **1000**

Parent 2: **0110**

After crossover: Offspring 1: **1000** = **8**

Offspring 2: **0110** = **6**

Role of Mutation

Mutation randomly changes one or more bits in a chromosome. It helps introduce new genetic variations, prevents premature convergence to local optimum solutions, increases population diversity, and improves the chances of finding the global optimum solution.

5 .During the training of a deep neural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model. (5 Marks)

Observed Trend

- Training accuracy increases from **60% to 95%**.
- Validation accuracy increases from **58% to 72%**, then decreases to **68%**.

Issue Identified

The model is experiencing **Overfitting**.

Reason

Initially, the model learns useful patterns from the training data. After several epochs, it starts memorizing the training data instead of learning general features. As a result, training accuracy continues to improve, while validation accuracy decreases because the model performs poorly on unseen data.

Techniques to Reduce Overfitting

1. **Early Stopping** – Stop training when validation accuracy stops improving.
2. **Dropout Regularization** – Randomly deactivate neurons during training to prevent overfitting.

Significance of Validation Accuracy

Validation accuracy measures how well the trained model performs on unseen data. It helps evaluate the model's generalization ability, detect overfitting, compare different models, and select the best-performing model for deployment.

6. A perceptron model is used to classify whether a loan is approved based on two features: income and credit score. The weights are 0.7 and 0.5, and the bias is –0.6. For an applicant with income = 1.5 and credit score = 2, compute the net input and apply the step activation function. Determine whether the loan is approved or rejected. (5 Marks)

Given:

- Weight w1=0.7w1=0.7
- Weight w2=0.5w2=0.5

- Bias b=−0.6b=−0.6
- Income x1=1.5x1=1.5
- Credit Score x2=2x2=2

Step 1: Calculate Net Input

Formula:

Net Input = $(w_1 \times x_1) + (w_2 \times x_2) + b$

Calculation:

Net Input = (0.7 × 1.5) + (0.5 × 2) – 0.6

= 1.05 + 1.00 – 0.60

= **1.45**

Step 2: Apply Step Activation Function

Step Function:

- If Net Input ≥ 0, Output = 1
- If Net Input < 0, Output = 0

Since **1.45 > 0**,

Output = 1

Result

Loan Approved.

7. In a neural network, a single neuron uses the delta learning rule. The initial weight is 0.4, learning rate is 0.2, and input is 3. The actual output is 0.5, while the target output is 0.9. Calculate the error, determine the weight update, and compute the new weight after one iteration. (5 Marks)

Given:

- Initial Weight (w) = 0.4
- Learning Rate (η) = 0.2
- Input (x) = 3
- Actual Output = 0.5
- Target Output = 0.9

Step 1: Calculate Error

Formula:

Error = Target Output − Actual Output

= 0.9 − 0.5

= **0.4**

Step 2: Calculate Weight Update

Formula:

$\Delta w = \eta \times \text{Error} \times \text{Input}$

$= 0.2 \times 0.4 \times 3$

= **0.24**

Step 3: Calculate New Weight

Formula:

New Weight = Old Weight + Δw

= 0.4 + 0.24

= **0.64**

Answer

- Error = **0.4**
- Weight Update = **0.24**
- New Weight = **0.64**

8. A classification model produces the following confusion matrix values: True Positives = 120, True Negatives = 200, False Positives = 30, and False Negatives = 50. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on these results, evaluate whether the model is suitable for applications where false negatives are critical. (5 Marks)

Given:

- True Positive (TP) = 120
- True Negative (TN) = 200
- False Positive (FP) = 30
- False Negative (FN) = 50

1. Accuracy

Formula:

Accuracy = (TP + TN) / (TP + TN + FP + FN) × 100

= (120 + 200) / (120 + 200 + 30 + 50) × 100

= 320 / 400 × 100

= **80%**

2. Precision

Formula:

Precision = TP / (TP + FP)

= 120 / (120 + 30)

= 120 / 150

= **0.80 = 80%**

3. Recall (Sensitivity)

Formula:

Recall = TP / (TP + FN)

= 120 / (120 + 50)

= 120 / 170

= **0.7059 = 70.59%**

4. Specificity

Formula:

Specificity = TN / (TN + FP)

= 200 / (200 + 30)

= 200 / 230

= **0.8696 = 86.96%**

5. F1-Score

Formula:

F1-Score = 2 × (Precision × Recall) / (Precision + Recall)

= 2 × (0.80 × 0.7059) / (0.80 + 0.7059)

= 1.1294 / 1.5059

= **0.75 = 75%**

Evaluation

The model has **80% accuracy** and **75% F1-score**, but the **Recall (70.59%)** indicates that **50 positive cases are missed (False Negatives)**. For applications such as medical diagnosis or fraud detection, where false negatives are critical, this model is **not ideal**. Improving recall by reducing false negatives would make the model more suitable for such applications

9. A Genetic Algorithm is used to maximize the function f(x) = 2x + 3 for an initial population of values {1, 3, 5, 7}. Compute the fitness of each individual and select the top two candidates. Perform a single-point crossover to generate offspring and explain how mutation can help avoid premature convergence. (5 Marks)

Given:
f(x) = 2x + 3
Population = {1, 3, 5, 7}

Fitness Values

x	Fitness
1	5
3	9
5	13
7	17

Best Candidates: 7 and 5

Single-Point Crossover

7 = **111**
5 = **101**

After crossover:

- Offspring 1 = **101 (5)**
- Offspring 2 = **111 (7)**

Mutation: Mutation changes random bits to increase diversity and avoid premature convergence.

10. During training of a neural network, the training loss decreases continuously, while validation loss decreases initially and then starts increasing after several epochs. Identify the issue occurring in the model. Explain the reason behind this behavior and suggest two techniques such as regularization or early stopping to improve generalization performance. (5 Marks)

Issue: Overfitting

Reason: The model memorizes training data, so validation loss increases after some epochs.

Solutions:

- Early Stopping
- Regularization (Dropout/L2)

Validation Loss: It helps measure the model's performance on unseen data.

11. A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is –0.4. For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not. (5 Marks)

Given:

- $w_1 = 0.5$, $w_2 = 0.3$
- $b = –0.4$
- $x_1 = 2$, $x_2 = 1$

Net Input = (0.5 × 2) + (0.3 × 1) – 0.4

= 1 + 0.3 – 0.4

= **0.9**

Since **0.9 ≥ 0**, Output = **1**

Result: Customer will purchase the product.

12. In a neural network, a neuron is trained using the delta learning rule. The initial weight is 0.6, the learning rate is 0.15, and the input is 2.5. The actual output is 0.7, while the target output is 1. Calculate the error, determine the weight update, and compute the new updated weight after one iteration. (5 Marks)

Given:

- Weight = 0.6
- Learning Rate = 0.15
- Input = 2.5
- Actual Output = 0.7
- Target Output = 1

Error = 1 – 0.7 = 0.3

Weight Update = 0.15 × 0.3 × 2.5 = 0.1125

New Weight = 0.6 + 0.1125 = 0.7125

Answer:

- Error = **0.3**
- Weight Update = **0.1125**
- New

13. A classification model produces the following confusion matrix values: True Positives = 140, True Negatives = 180, False Positives = 25, and False Negatives = 35. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on the results, analyze whether the model is suitable for applications such as fraud detection where false positives are costly. (5 Marks)

TP = 140, TN = 180, FP = 25, FN = 35

Accuracy

= (140 + 180) / (140 + 180 + 25 + 35) × 100

= **84.21%**

Precision

= 140 / (140 + 25)

= **84.85%**

Recall

= 140 / (140 + 35)

= **80%**

Specificity

= 180 / (180 + 25)

= **87.80%**

F1-Score

= 2 × (0.8485 × 0.80) / (0.8485 + 0.80)

= **82.38%**

Conclusion: The model is suitable for fraud detection because it has high precision (84.85%), resulting in fewer false positives.

14. A Genetic Algorithm is applied to maximize the function f(x) = 3x + 2 for an initial population {2, 5, 7, 9}. Compute the fitness value for each individual and select the

best candidates. Perform a crossover operation to generate new offspring and explain how mutation helps in maintaining diversity and avoiding local optima.
(5 Marks)

Given:
f(x) = 3x + 2
Population = {2, 5, 7, 9}

Fitness Values

x	Fitness
2	8
5	17
7	23
9	29

Best Candidates: 9 and 7

Crossover

9 = **1001**
7 = **0111**

After single-point crossover:

- Offspring 1 = **1001 (9)**
- Offspring 2 = **0111 (7)**

Mutation: Mutation changes random bits, increases diversity, and helps avoid local optimum solutions.

15. During the training of a deep learning model, the training accuracy increases from 65% to 97%, while validation accuracy increases from 63% to 75% and then fluctuates around 72%. Identify the issue present in the model. Explain why this happens and suggest techniques such as dropout and data augmentation to improve the model's performance. Also, discuss the importance of validation metrics in model evaluation. (5 Marks)

Issue: Overfitting

Reason: The model learns the training data too well and cannot generalize to new data, causing validation accuracy to stop improving.

Techniques to Improve Performance:

- Dropout
- Data Augmentation

Importance of Validation Metrics: Validation metrics measure the model's performance on unseen data, help detect overfitting, and assist in selecting the best model.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand